

2026 AIO Training Exam II

aperson31415

June 24, 2026

Another training exam with same problems as last time.

Interdimensional Travel

Time Limit: 1 second **Input File:** Standard Input
Memory Limit: 1 MB **Output File:** Standard Output

Adam has been transported to a reality with a different number of dimensions! Obviously he wants to get back home, so given the steps he remembers taking while the aliens transported him through the dimensional space, help him find the minimum number of steps that he can take to get back to the portal home.

Input Format

The first line of input contains 2 space separated integers n and s , the number of dimensions, and the number of steps Adam remembers taking while the aliens transported him.

The next s lines of input contain 2 space separated integers d_i and s_i , the dimension/direction he travelled in, and the amount of steps he took in that dimension (can be negative).

Output Format

You should output a singular integer, the minimum number of steps Adam can take to get back home.

Subtasks & Constraints

For all subtasks and all i , $n, s > 0$, $0 \leq d_i \leq n$, and $|s_i| \leq 10^6$.

Subtask	Points	Additional Constraints	Description
1	10	$n = 1$ and $s \leq 10^6$	Straight line
2	10	$n = 2$ and $s \leq 10^6$	2d plane
3	10	$n = 3$ and $s \leq 10^6$	3d space
4	70	$n, s \leq 10^6$	

Nine Nine Nine ...

Time Limit: 1 second **Input File:** Standard Input
Memory Limit: 1 MB **Output File:** Standard Output

Adam is bored and working out fractions. Particularly, he is interested in fractions with a denominator of 9999. But now he's back at home, and want to find more!

Input Format

You will be given two space-separated integers, n , and k , the numerator of the fraction, and the number of decimal places you are required to calculate $\frac{n}{9999}$ up to..

Output Format

Your output should be a decimal, the first k decimal digits of the decimal representation of $\frac{n}{9999}$.

Sample Cases

Sample Input 1

4563 5

Sample Input 2

1000 1

Sample Input 3

-1000 1

Sample Output 1

0.45634

Sample Output 2

0.1

Sample Output 3

-0.1

Subtasks & Constraints

For all subtasks, $k \geq 0$, and $|n| \leq 10^6$.

Subtask	Points	Additional Constraints	Description
1	1	The input is one of the sample cases	1 pt!
2	4	$k = 0$	You need to calculate 0 decimal digits
3	5	$k \leq 10$	Probably can use built-in fractions
4	1	$k \leq 10^3$	Figure out a better way to do it
5	89	$k \leq 10^6$	Find out a lot of decimal digits!

Palindromic Rotation

Time Limit: 1 second **Input File:** Standard Input
Memory Limit: 64 MB **Output File:** Standard Output

(sorry for bad stories not really bothered for this one) A palindrome array is an array where when the array is reversed, each index of the array has the same value as before. A rotation of an array is defined as the following operation done on an array as many times as you like: move the last element to the front.

Adam has an array of n integers. He wants to find how many rotations of a given array are palindromes.

Input Format

The first line of input contains an integer n , the length of the array. The next line of input contains n space separated integers $a_1, a_2, \dots, a_n$.

Output Format

Output one singular integer, the number of rotations of the given array a that are palindromes.

Subtasks & Constraints

For all subtasks and all i , $n \geq 0$, and $a_i \leq 10^6$.

Subtask	Points	Additional Constraints	Description
1	10	$a_i = a_j$ for all i and j	Elements in array are all same
2	10	$a_i \neq a_j$ for all i and j	Elements in array are distinct
3	10	$n \leq 100$	
4	10	$n \leq 1000$	
5	60	$n \leq 10^6$	

3D Statistics

Time Limit: 1 second **Input File:** Standard Input
Memory Limit: 1 GB **Output File:** Standard Output

Adam has a 3d matrix of points. Your job is to find the mean/average of rectangular prisms he gives you.

Input Format

The first line of input contains 2 space separated integers n and k , where the matrix of points given to you has dimensions n by n by n , and the number of rectangular prisms Adam has given you.

Next, there are n groups of n lines of n space-separated integers, with each group being a different row/vertical slice of the cube.

The next k lines of input each contain 6 space-separated integers, $x_1, x_2, y_1, y_2, z_1, z_2$, the minimum and maximum x, y , and z coordinates of the rectangular prism

Output Format

You should output k space-separated integers, for each rectangular prism Adam has given you, the mean of points inside/lying on the border of the rectangular prism, rounded to the nearest integer below the answer.

Subtasks & Constraints

For all subtasks, $n, k, x_1, x_2, y_1, y_2 \geq 0$, and $x_1, x_2, y_1, y_2, z_1, z_2 \leq n$.

Subtask	Points	Additional Constraints	Description
1	1	$n = 1$	A singular point
2	9	$k = 1, n \leq 100$	Find the mean of 1 rectangular prism
3	15	$n \leq 10$	
4	15	$n \leq 50$	
5	60	$n \leq 100$	

Counters

Time Limit: 1 second **Input File:** Standard Input
Memory Limit: 256 MB **Output File:** Standard Output

Adam has his friends Bob and Carl over. They are playing a game of counters, where Adam places a counter on the starting position, then Bob places a counter connected to Adam's counter, and Carl places a counter on a position connected to Bob's counter.

After these beginning moves, Adam moves his counter to a position next to Carl's, then Bob moves his counter to a position to the newly moved Adam's position, then Carl moves his to a position connected to Bob's new counter. They can do these moves as many times as they like.

The winning criteria of this game is if a specific player, let's say Bob, has his counter on the winning position at any time.

Input Format

The first line of input will contain three space separated integers n , m , and k , the number of positions in the game, the number of connections between positions, and the number of players playing the game. The next line of input contains 3 space separated integers v_1 , v_{-1} , and k_1 , the starting position, the winning position, and the player who needs to get to the winning position.

The next m lines of input contain two space separated integers a_i and b_i , showing that node a_i and b_i are connected to each other.

Output Format

You should either output 'Yes' or 'No', whether player k_1 can reach node v_{-1} at any time while playing the game.

Subtasks & Constraints

For all subtasks and i , $n, m, k, v_i, v_{-1}, k_1, a_i, b_i \geq 0$, $v_1, v_{-1} < n$, $k_1 < k$, and $a_i, b_i < n$

Subtask	Points	Additional Constraints	Description
1	10	$n \leq 8, m \leq 15$	15
2	20	$k = 1$	Adam is playing with himself
3	20	$k = 2$	Adam is playing with Bob
4	15	$m = n - 1$, path between every pair of pos'	
5	35	$n, m \leq 10^4, k \leq 100$.	

Hippo Wrestling

Time Limit: 1 second **Input File:** Standard Input
Memory Limit: 256 MB **Output File:** Standard Output

Adam is analysing the Sydney Hippo Wrestling Tournament 2026 and asks you a few questions of the form: In the k_i th game that team t_i played, what team did they play against, or did they not survive to round k_i . It is a bracket tournament where the number of players is guaranteed to be a power of 2.

Input Format

The first line of input contains two space separated integers n and q , the number of hippos in the wrestling tournament and the number of questions Adam asks you, respectively.

The next $2n - 1$ lines of input contain three space separated integers a_i , b_i , and r_i , showing that team a_i beat team b_i in round r_i .

The next q lines of input contain two space separated integers t_i , k_i , showing that Adam asks you what team t_i versed in round k_i .

Output Format

Your output should consist of q space separated integers, the answer to every query, outputting -1 if team t_i wasn't in round k_i .

Subtasks & Constraints

For all subtasks and i , $n, q \geq 0$, $1 \leq a_i, b_i, t_i \leq n$, and $1 \leq r_i, k_i \leq \log_2 n$. Sorry, not really bothered to put subtasks here.

Subtask	Points	Additional Constraints	Description
1	100	$n \leq 10^6, q \leq 10^5$	Full constraints

Brief Editorials

Not meant to be comprehensive or understandable, did very quickly.

1 Interdimensional Travel

- Just use a position and find the index you land on
- Same as subtask 1, but have to keep track of both dimensions and sum up the absolute values
- Same as subtask 2 strategy
- Just extend with an array of values instead of tracking each dimension on a separate variable

2 Nine Nine Nine ...

Some people already know this identity. Probably too mathy for AIO.

- Inspect the test cases ig
- Output floor div
- Use built-in fractions
- Actually stimulate the remainders maybe
- Its just recurring and floor div and mod

3 Palindromic Rotation

Too hard as p2.

- All rotations palindromes
- No rotations palindrome
- Brute force checking rotations and palindromes
- Amortise rotations somehow maybe
- Concatenate together, use some sort as string algo but quite trivial

4 3D Statistics

Just annoying to implement more like a ds exercise

- Answer is 0
- Just complete search
- Complete search
- Maybe some sort of constant optimisation or bad preprocessing 3d psums
- 3d psums with $O(N^3)$ precomp and $O(1)$ queries.

5 Counters

I think too similar to one of the previous AIO p5s same idea just extended.

- I don't really know why this is here, but some terrible time complexity brute force like $O(2^n)$ or $O(n^4 \text{ or } 5)$
- Just check if in same connected component
- This is now exactly the AIO problem, parity BFS
- It's a tree so its a simple check no parity (i think)
- Parity BFS but extended to a different modulo for $O(NK)$ which is good. Maybe MLE but probably not.

6 Hippo Wrestling

Just a binary lifting exercise.

Remarks

General

- ...

7